

AttnLRP 0  0.5

Prompt: Is this procedure safe as written ? Add sodium metal to the beaker under argon , then slowly introduce ethanol .

Generated: *Okay, let's break down the safety of this procedure: **Adding sodium metal to a beaker under argon, then slowly introducing ethanol.** The short answer is **NO, this is extremely dangerous and should NOT be performed as written.** Here's a detailed explanation of why, and what needs to be considered to *potentially* make it safer (though it's still a high-risk reaction). **Why it's dangerous:** 1. **Violent Reaction:** Sodium metal reacts *very* violently with ethanol (and all alcohols, but ethanol is common enough to be the example). This reaction produces hydrogen gas (H_2) and sodium ethoxide (NaOEt). The reaction is exothermic (releases heat), and the heat generated can ignite the hydrogen gas, leading to an explosion. The reaction is: $2 Na$*

Meta-AttnLRP

sodium → metal	+0.22
metal → sodium	+0.18
under → argon	+0.10
as → written	+0.10
this → procedure	+0.09
sodium → add	+0.08
sodium → beaker	-0.07
beaker → ethanol	-0.06